

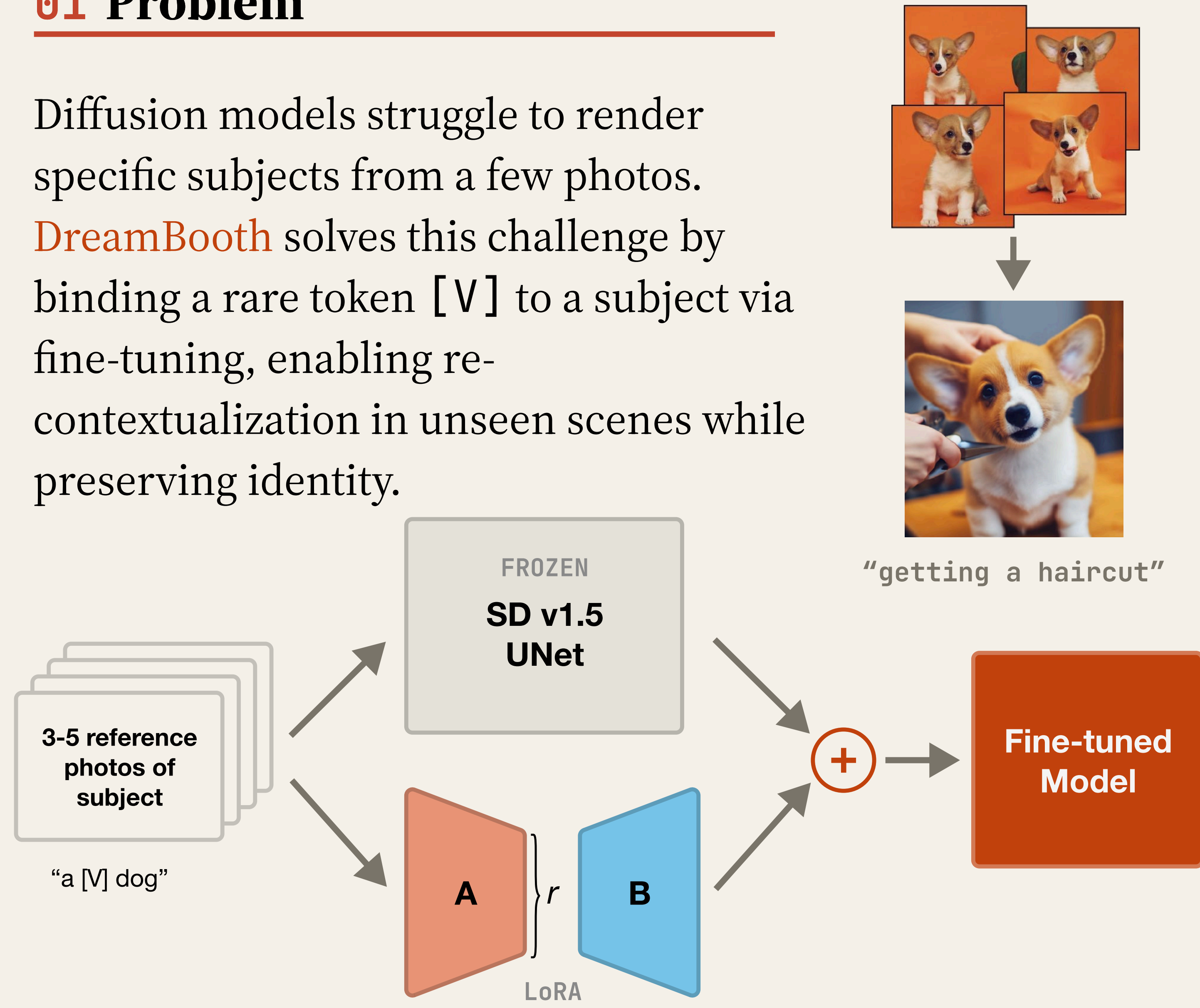
# DreamBooth + LoRA + VideoGen

Fine-tuning text-to-image diffusion models for subject-driven generation

Andrew Noviello, Eashan Vagish, Ronald Leung, Sean Lee  
Cornell University

## 01 Problem

Diffusion models struggle to render specific subjects from a few photos. **DreamBooth** solves this challenge by binding a rare token [V] to a subject via fine-tuning, enabling re-contextualization in unseen scenes while preserving identity.



### 1 Diffusion loss ( $\mathbf{x}_\theta$ form)

$$\mathcal{L}_{\text{diff}} = w_t \left\| \hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x} + \sigma_t \epsilon, \mathbf{c}) - \mathbf{x} \right\|_2^2$$

Trains the model to recover the clean image  $\mathbf{x}$  from noised input.

$\mathbf{x}$ : clean image (subject image)  
 $\mathbf{c}$ : text conditioning (e.g., "a [V] dog")

$\epsilon \sim \mathcal{N}(0, \mathbf{I})$ : Gaussian noise  
 $w_t$ : time-dependent weighting

### 2 Class-specific prior preservation loss (PPL)

$$\mathcal{L}_{\text{PPL}} = w'_t \left\| \hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x}_{\text{pr}} + \sigma_t \epsilon', \mathbf{c}_{\text{pr}}) - \mathbf{x}_{\text{pr}} \right\|_2^2$$

Preserves the model's ability to generate images of the same class (e.g., "a dog").

$\mathbf{x}_{\text{pr}}$ : sample image from the prior dataset  
 $\mathbf{c}_{\text{pr}}$ : corresponding caption (e.g., "a dog")

$\epsilon' \sim \mathcal{N}(0, \mathbf{I})$ : Gaussian noise  
 $w'_t$ : time-dependent weighting

### 3 Final DreamBooth objective (expectation form)

Balances subject learning and prior preservation.

$$\mathbb{E}_{\mathbf{x}, \mathbf{c}, \epsilon', t} \left[ w_t \left\| \hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x} + \sigma_t \epsilon, \mathbf{c}) - \mathbf{x} \right\|_2^2 + \lambda w'_t \left\| \hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x}_{\text{pr}} + \sigma_t \epsilon', \mathbf{c}_{\text{pr}}) - \mathbf{x}_{\text{pr}} \right\|_2^2 \right]$$

$\lambda$ : trade-off weight between the two losses

$\mathbb{E}_{\mathbf{x}, \mathbf{c}, \epsilon', t}$ : expectation over data, captions, noise, and timesteps  $t$

Diffusion loss

$$w_t \left\| \hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x} + \sigma_t \epsilon, \mathbf{c}) - \mathbf{x} \right\|_2^2$$

+

$\lambda \cdot \text{PPL}$

$$w'_t \left\| \hat{\mathbf{x}}_\theta(\alpha_t \mathbf{x}_{\text{pr}} + \sigma_t \epsilon', \mathbf{c}_{\text{pr}}) - \mathbf{x}_{\text{pr}} \right\|_2^2$$

↓

Optimize

### 4 Extension: training with LoRA

$$h = W\mathbf{x} + \frac{\alpha}{r} B A \mathbf{x} \quad \text{where } W \in \mathbb{R}^{d \times k}, A \in \mathbb{R}^{r \times k}, B \in \mathbb{R}^{d \times r}, r \ll \min(d, k)$$

Only the low-rank matrices  $A, B$  (and scalar  $\alpha$ ) are updated, keeping training light.

## 03 Generated Images

4 TRAINING IMAGES



FINE-TUNE • BIND

“a [V] dog”

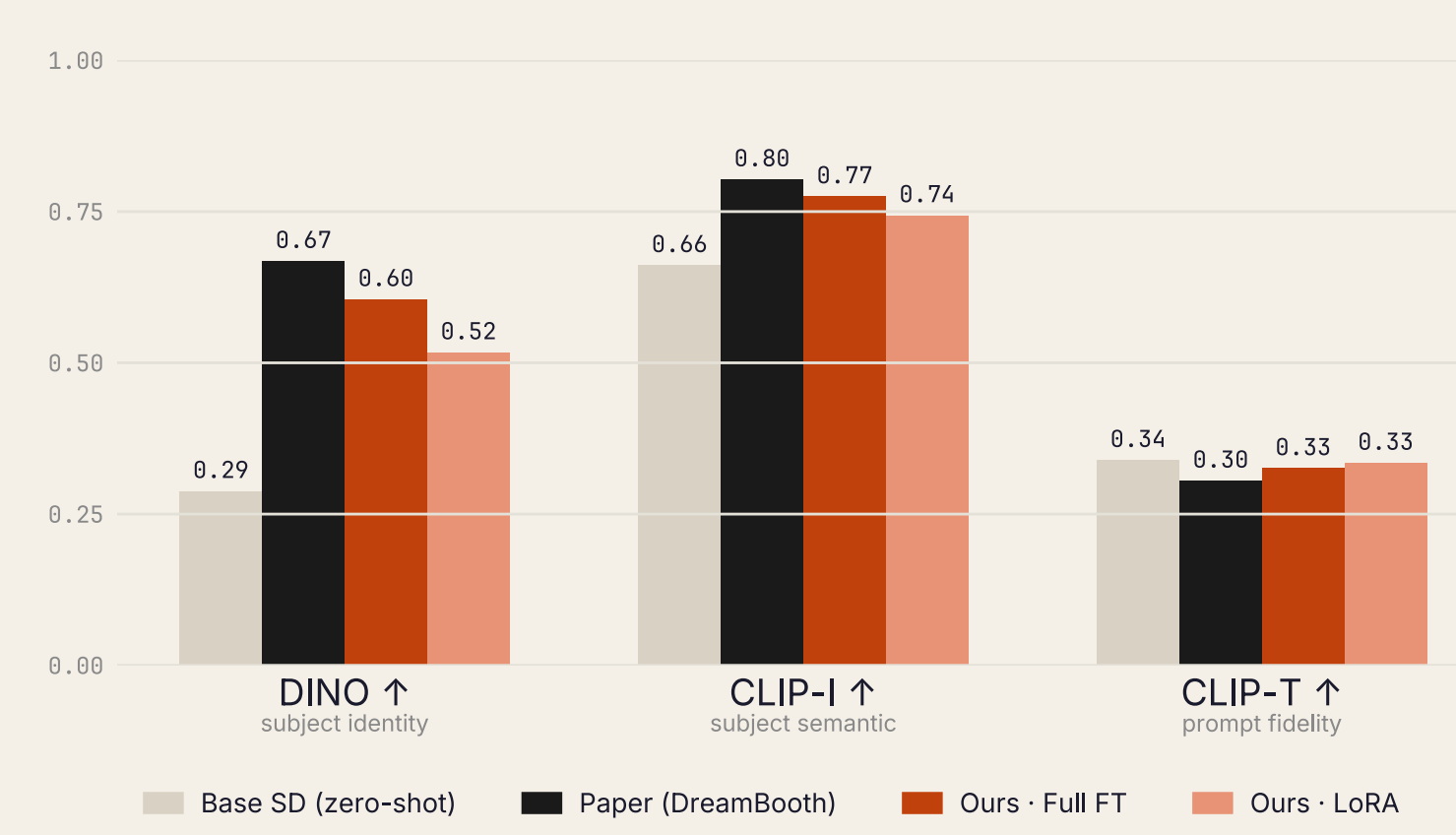
rare token bound to subject

NOVEL CONTEXTS • SAME SUBJECT



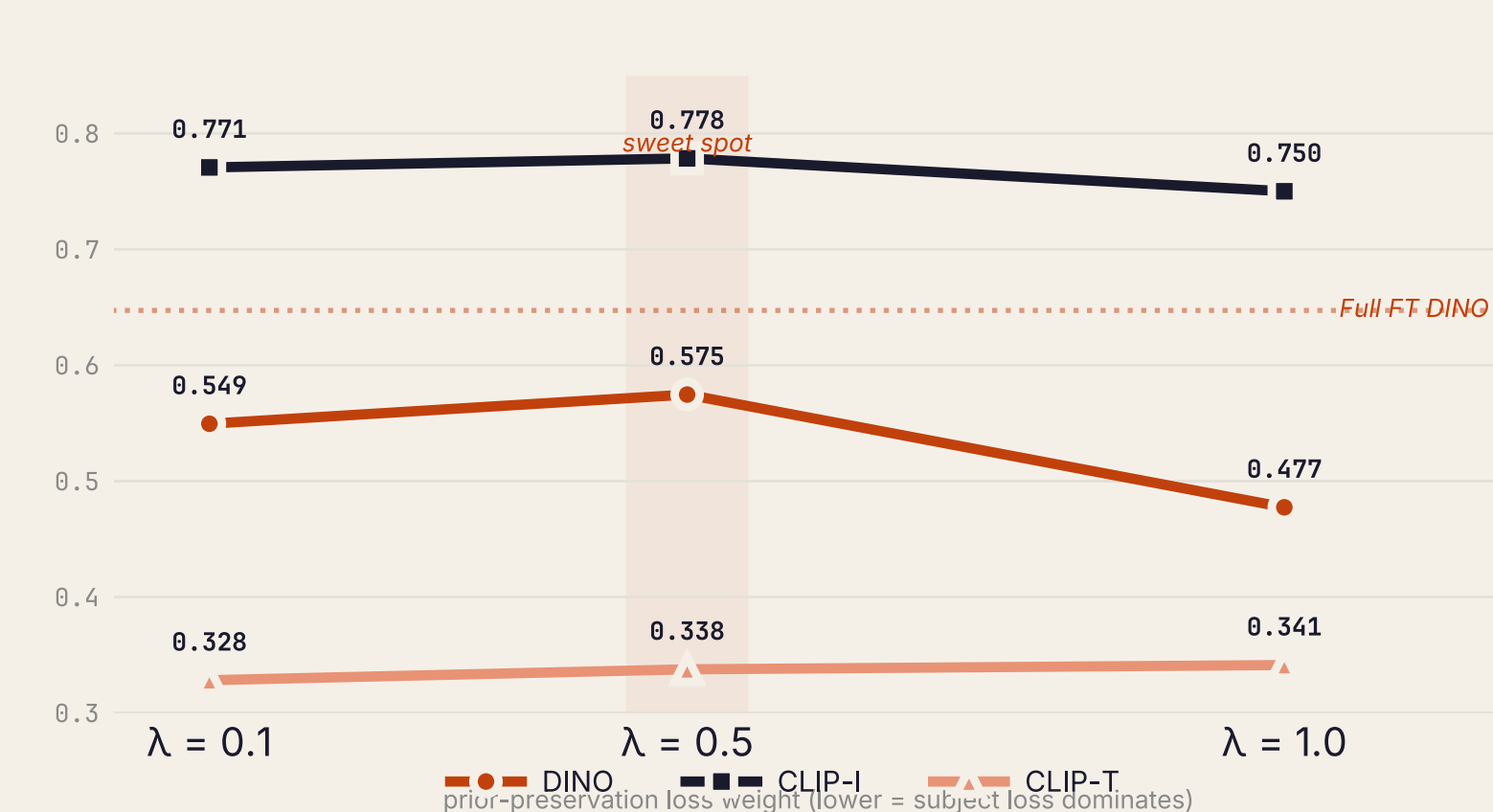
## 04 Results & Discussion

Subject and prompt fidelity vs the paper



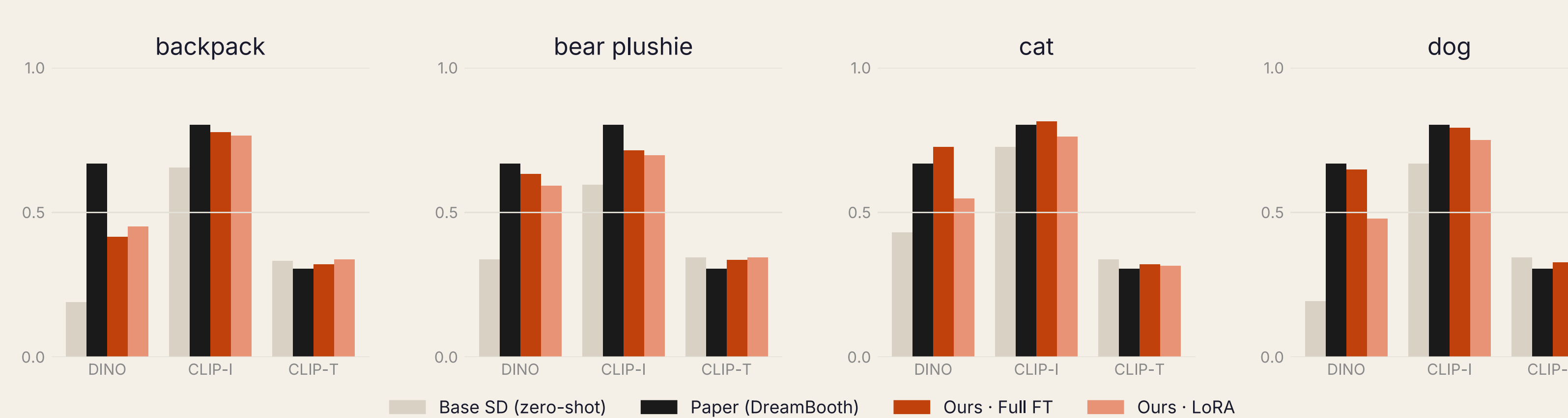
Our Full FT matches the paper on subject semantics (CLIP-I 0.77 vs 0.80) and exceeds it on prompt fidelity (CLIP-T 0.33 vs 0.30). DINO sits 0.07 below the paper's 30-subject average, within their reported per-subject variance. LoRA recovers ~86% of Full FT's DINO at <1% of trainable parameters.

Prior-loss weight • LoRA ablation on dog



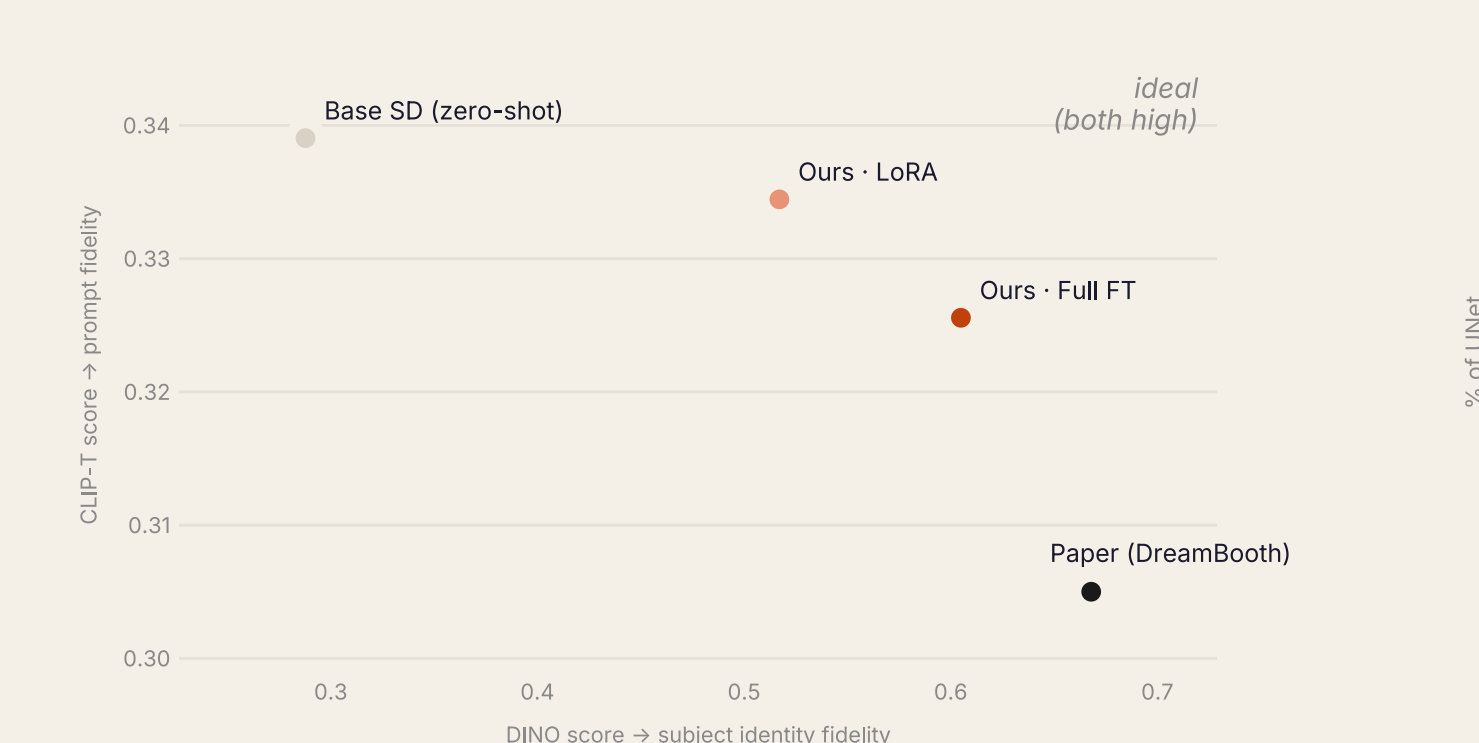
Both subject metrics peak at  $\lambda = 0.5$  — the LoRA-specific sweet spot.  $\lambda = 1.0$  over-regularizes (subject under-fit);  $\lambda = 0.1$  starts to drift on prompts. Validates the paper's claim that prior preservation regularizes against language drift, while showing LoRA prefers a weaker prior than Full DreamBooth.

Per-subject metric breakdown



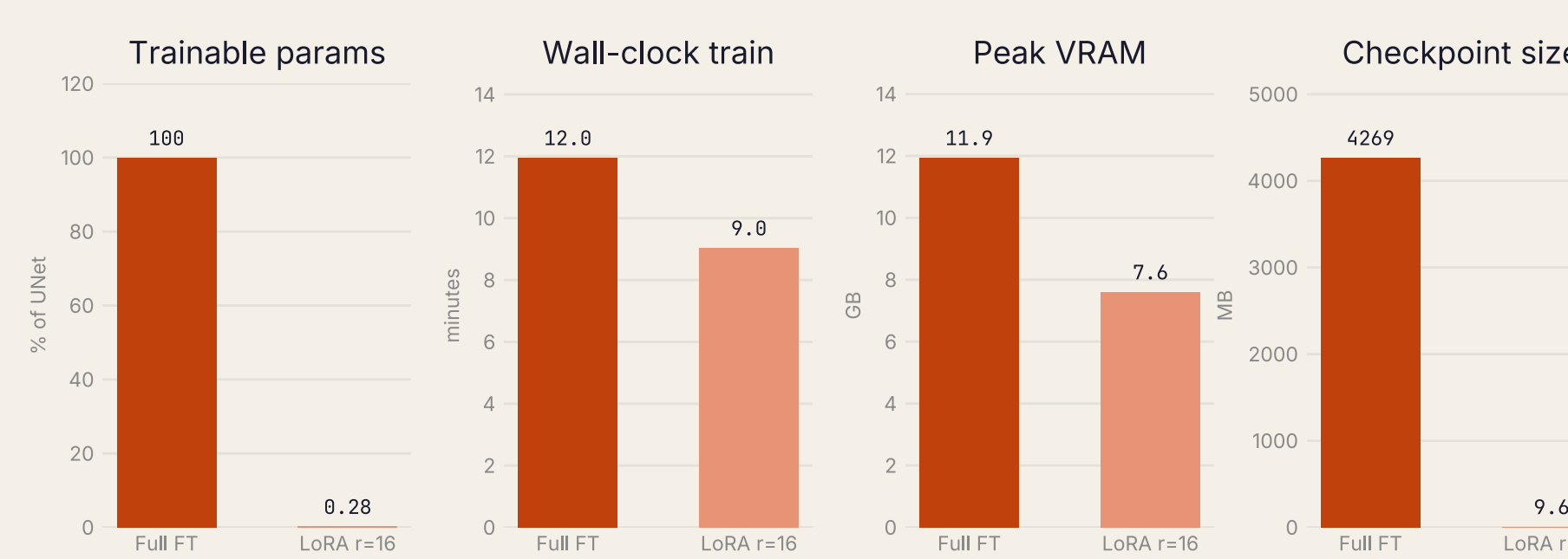
Cat reaches DINO 0.73 (above paper's 0.67 average) and dog 0.65; backpack lags at 0.42, consistent with the paper's note that generic object classes are the hardest for instance-level identity. CLIP-I and CLIP-T stay stable across all four subjects — the per-subject variance lives almost entirely in DINO.

The subject-prompt fidelity trade-off



DreamBooth's central tension: aggressive fine-tuning improves subject identity (DINO ↑) but risks language drift on prompts (CLIP-T ↓). Both our variants land near the paper's Pareto point, far from base SD's near-zero subject identity.

LoRA • same job, fraction of the cost



LoRA trains 357x fewer parameters (0.28% of the UNet) and produces checkpoints 445x smaller (4.3 GB → 9.6 MB) while recovering 86% of Full FT's DINO and matching its CLIP-T. Wall-clock and VRAM savings are modest (25% / 36%) since the frozen UNet still runs forward each step. Single RTX 4090, 1000 steps, batch 1, fp32.

\*We also ran experiments with three additional base models: SDXL, CommonCanvas, and FLUX-2 4B. FLUX-2 4B (a rectified flow transformer) required us to adapt the DreamBooth PPL for the flow matching objective.

## 05 Beyond LoRA: More Extensions!

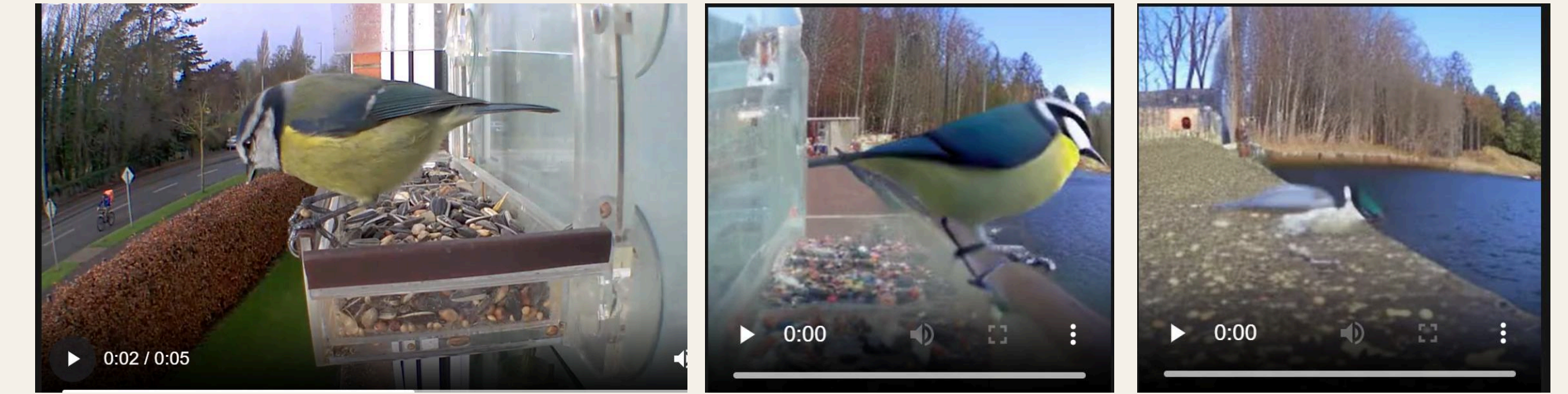
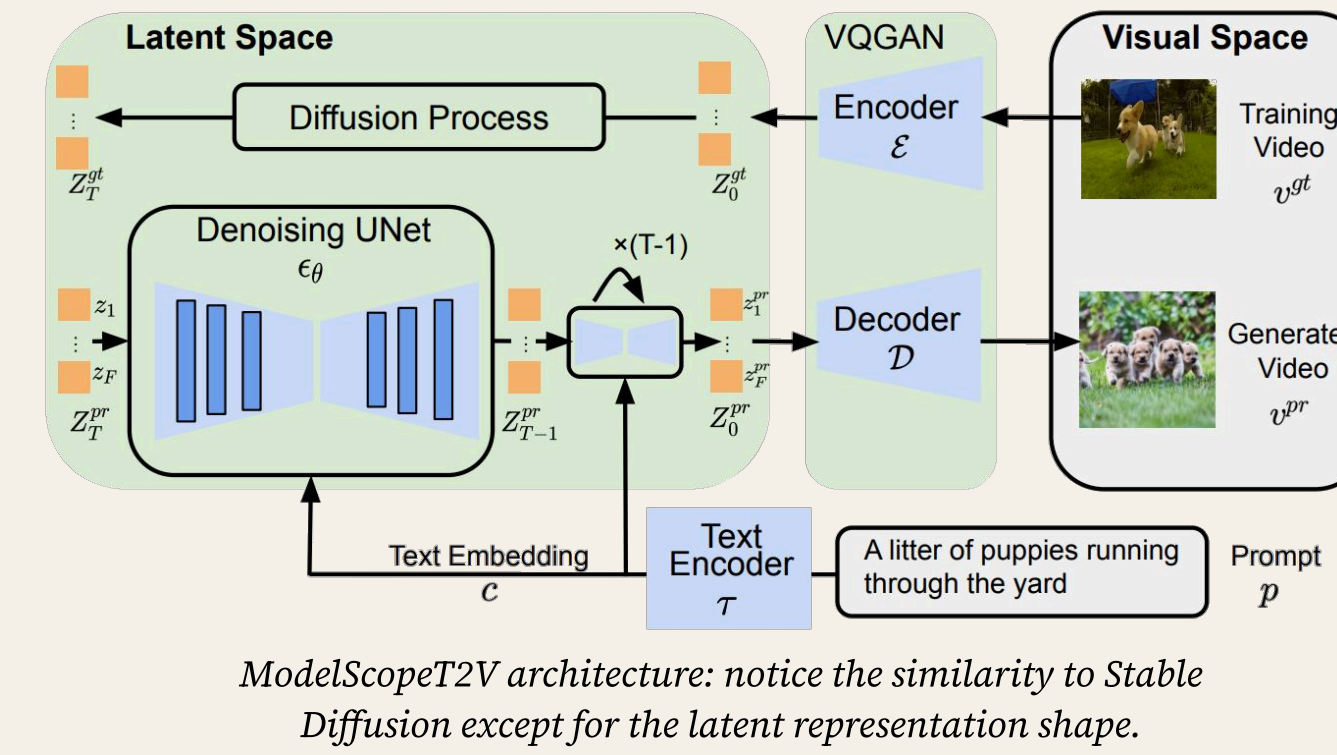
Video Generation: DreamBooth on ModelScopeT2V

ModelScopeT2V: text-to-video latent diffusion model similar to Stable Diffusion

Clips represented as sequences of compressed frame latents; motion across frames captured by spatio-temporal UNet

DreamBooth objective easily adapted to video base model: diffusion loss computed normally for the new latent shapes & prior preservation loss from sample dataset video clip priors

Result: achieved strong subject binding even in the higher dimensional latent video problem using only 5 short input video clips. \*Note that, due to compute resources, we only finetuned ModelScopeT2V with LoRA.



Left: Frame captured from a training video. Middle: Frame captured from a prompt "[V] bird with lake". Right: Frame captured from a prompt "bird with lake". Shows that the rare token binds to the subject, even in text-to-video settings.

## Post-training is Necessary for Personalization: Classifier Guidance Fails

Are there alternatives to finetuning for personalization? We wanted to check whether we could achieve similar performance to DreamBooth without weight-level updates, using classifier guidance. For guidance, we added an auxiliary L2 loss between the current noisy latent and a target latent, forward-diffused to the same noise level via the scheduler.

We tried three target-selection strategies:

- the pooled mean of all subject-image latents
- a randomly chosen single dataset latent fixed per sample
- a freshly resampled dataset latent at every denoising step.

We varied the auxiliary loss weight and the guidance weight schedule for testing.

Guidance largely failed, resulting in either blurriness or mode collapse. This experiment made us appreciate DreamBooth even more!



Example of Mode Collapse: Ignores Prompt



Examples of Blurriness Due to Latent Pooling

## 06 Conclusion & Future Work

### What we did

- Re-implemented DreamBooth architecture
- Roughly matched original paper performance on 4 subjects
- Implemented LoRA and showed it worked for DreamBooth
- Experimented with different LDMs
- Generalized DreamBooth to videos
- Tested classifier guidance as a potential alternative to finetuning



### Future Work

- Train model on all subjects
- Find more rare tokens and train on multiple subjects at the same time
- Find optimal rank of matrix used for LoRA
- Utilize technique for generation of subject-specific videos with larger video models
- Test whether DreamBooth formulation can generalize to multiple subjects

